

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and implemented **functional verification environments** and **test plans** for enterprise Java/Spring Boot microservices at scale, designing **testbenches** and **test cases** that validated complex business logic across payment-processing and ERP integration systems, achieving **99.9% SLA compliance** in production deployments.
- Engineered **random test generators**, **bus functional models (BFMs)**, and **functional constraints** using **Python** and **Shell scripting** to automate data validation pipelines and ETL workflows, generating **50+ daily automated test suites** for production support monitoring via **Splunk**, **Dynatrace**, and **CloudWatch**.
- Implemented **assertion-based verification** and **checkers** across interconnected **SoC-based** payment systems and ERP modules, collaborating with cross-functional design engineers and business architects to develop **coverpoints** and validation **scoreboards** that reduced post-deployment defects by **35%**.
- Designed and executed **formal verification techniques** and validation lifecycle management for mission-critical SWIFT messaging and ISO 20022 payment-flow components, leveraging **Jenkins CI/CD pipelines**, **Git** version control, and change-control methodologies to ensure regulatory compliance and zero-downtime system transitions.
- Developed high-level **transactional models** and verification infrastructure using **Python**, **SQL (PostgreSQL, MySQL, Oracle)**, and **Docker containerization** to validate **memory controllers**, data-flow integrity, and cross-system AMBA protocol interactions across payment-operations dashboards and ERP platforms, reducing test-cycle duration by **40%**.
- Led problem-solving and root-cause analysis in fast-paced, innovative production environments using **Linux** log analysis, **TCL scripting**, and **AWS (EC2, Lambda)** infrastructure to identify and resolve critical defects in pre-silicon and post-implementation validation phases, mentoring junior engineers on formal verification best practices and documentation standards.

SKILLS

Pre-Silicon Verification & Formal Methods – System Verilog Assertions (SVA), formal verification techniques, assertion-based verification, functional verification, testbenches, test plans, test cases, random test generators, bus functional models (BFMs), checkers, scoreboards, coverpoints, covergroups.

Verification Tools & Methodologies – UVM simulation-based validation, verification environments, validation lifecycle management, industry standard verification tools, change-control management, test automation frameworks.

Digital Design & SoC Architecture – SoC-based CPUs, NoCs, AMBA protocols, memory controllers, complex digital design blocks, pre-silicon validation for ASIC/SoC/FPGA designs.

Programming & Scripting Languages – Python, Java, SQL (PostgreSQL, MySQL, Oracle), Shell scripting, TCL, C/C++, REST APIs.

Cloud, DevOps & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD pipelines, Git branching, Linux system administration, log analysis.

Monitoring, Analytics & BI Tools – Splunk, Dynatrace, CloudWatch, Power BI, Tableau, data modelling, ETL pipelines.

Domain & Regulatory Knowledge – Payment systems, SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA/cross-border payments, PSD2, GDPR compliance, ERP integration, SoC architectural patterns.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

NPU AI Accelerator Pre-Silicon Validation Framework (HCL Technologies)

- Architected and deployed a comprehensive **functional and formal verification environment** for a multi-block **AI accelerator SoC**, implementing **System Verilog Assertions**, **testbenches**, and **UVM simulation-based validation** frameworks that validated compute datapaths, memory controllers, and NoC interconnects across **250+ test cases**.
- Developed **random test generators**, **bus functional models (BFMs)** for **AMBA protocol** compliance, and assertion-based **checkers** using **System Verilog** and **Python** to detect protocol violations and functional regressions, achieving **98% code coverage** and reducing time-to-silicon by **6 weeks**.
- Engineered **formal verification techniques** and **transactional models** to validate critical **memory controller** and datapath functionality, creating **covergroups** and **scoreboards** that measured functional completeness across **12 major design blocks** and coordinated sign-off with design and AI architecture teams.
- Implemented **test plan development**, validation **lifecycle management**, and **Jenkins CI/CD automation** pipelines using **TCL scripting** and **Shell** to orchestrate **50+ daily regression test suites**, integrated with **Git** version control and comprehensive defect tracking via **Jira**, enabling rapid iteration and **zero-slip tape-out schedules**.
- Collaborated across verification engineers, design engineers, and AI architects through **Agile/Scrum ceremonies** and **Confluence documentation** to translate architectural requirements into formal verification constraints, mentoring junior engineers on formal verification methodology and assertion design patterns.

Payment-Operations Pre-Silicon Data-Integrity Validation Suite (HCL Technologies)

- Built a specialized **functional verification testbench** and **test case suite** for SWIFT payment-flow components processing **ISO 20022 message formats**, validating transaction atomicity, settlement reconciliation, and **SoC-based** datapath correctness across payment-operations dashboards using **formal verification** techniques and **assertion-based checkers**.
- Designed **bus functional models (BFMs)** and **transactional models** using **Python** and **SQL (PostgreSQL, Oracle)** to simulate cross-border payment flows and SEPA/instant-payment processing scenarios, enabling **test case generation** that validated regulatory compliance for **GDPR data-protection** requirements and **PSD2 open-banking** standards.
- Implemented **covergroups**, **scoreboards**, and **System Verilog Assertions** to verify memory-coherency, AMBA protocol interactions, and data-integrity constraints, reducing post-deployment defects in payment-critical systems by **42%** and achieving **96% functional coverage** across all message-type validation paths.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)

- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)